Xi'an Jiaotong-Liverpool University

西交利物浦大学

MODULE CODE	EXAMINER	DEPARTMENT	TEL
CPT101		COMPUTER SCIENCE	

1st SEMESTER 2022/23 END TERM EXAMINATIONS

BACHELOR DEGREE – Year 2

COMPUTER SYSTEMS

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. **This is an open-book exam.**
2. **Answer all questions.**
3. **Answers should be written in the answer booklet(s) provided.**
4. **No calculator is allowed during the examination.**
5. **Only answers in English are accepted.**

THIS PAPER MUST NOT BE REMOVED FROM THE EXAM HALL.

28. Supply correct arguments and/or instructions to the missing places **(i)** to **(x)** in the following bubble sort program segment that compares two integers and swap them, resulting in a sorted list of integers in ascending order. **(Total 10 marks: each worth 1 mark)**

```
int MAX_SZ = 4
int intArray[(i)] = {6, 1, 8, 4};
(ii){
  (iii) esi, (iv)     ; load first item's memory address
  mov   eax, (v)      ; move content of memory address
  (vi)    [esi+(vii)], eax ; compare adjacent items
  (viii)  L3        ; jump out to L3
  (ix) eax, [esi+4]   ; otherwise swap contents
  (x) [esi], eax  ; continue swapping
}
```